Supplementary table 1 *Comparison of the pilot version of ZWA versus the evolved ViRAE version in terms of software features as well as de novo assembly performance based on the Anopheles sacharovi mosquito RNA-seq sample (SRA accession number SRR13449040), its 28S and 18S rRNAs given as reference (GenBank accession numbers MT808434 & MT808462), and Xanthi rhabdovirus (GenBank accession number MW520377) genome used for testing.*

	Software features							***de novo* assembly performance**	
	Implementation	Input file type	Prerequisites	Alignment software	*De novo* assembly	User adjustability	Execution time (seconds)	Final quality score (%)	Max query covbases (%)
ZWA	Bash script	FASTQ (single-end) & FASTA (reference)	7 (BIOAWK, BLAST+, BBMAP reformat.sh, faSomeRecords, BWA, samtools, Trinity)	BWA & BLASTn	Integrated (Trinity required)	None	1006344.582	63.00	24.71
ViRAE	Bash script & Web platform	FASTQ (single- OR paired-end) & FASTA (reference) OR BAM only (alignment)	3 (Nextflow, BWA, samtools)	BWA only	Optional (MEGAHIT recommended)	Alignment stringency	228.838	66.10	34.85